

Research paper

Biopolymers can be used for consolidation of lunar and Martian regoliths

Nitin Gupta^{a,b}, Alope Kumar^a, Koushik Viswanathan^a^a Department of Mechanical Engineering, Indian Institute of Science, Bangalore, 560012, Karnataka, India^b Université Paris-Saclay, CEA, CNRS, SPEC, 91191, Gif-sur-Yvette, France

ARTICLE INFO

Keywords:

Lunar regolith
Martian regolith
Space habitation
Extraterrestrial construction
Biopolymers
Regolith simulant

ABSTRACT

Creating long-term extraterrestrial habitats to support continuous human occupation requires structures that can bear load, while remaining resilient to large temperature fluctuations. Exploiting locally available regolith is part of a paradigm termed *in situ* resource utilization (ISRU) that advocates minimal dependence on terrestrial supplies via local resource exploitation for construction. In this work, we report on a consolidation method for making load-bearing structures out of lunar and Martian regolith simulants using biopolymer binders. We quantify the compressive and tensile strength of the final brick-like structures, termed biopolymer regolith composites, and investigate their dependence on biopolymer concentration and chemistry. A pressure-assisted die casting (PADC) process is developed that can produce bricks with compressive strengths of ~15 MPa and tensile strengths of around 2 MPa, from regolith simulants. We finally report on the performance of these bricks under large temperature and pressure variations and find that their strengths remain unchanged. We believe that this makes biopolymer regolith composite bricks suitable candidates for use in harsh extraterrestrial environments, while adhering to the guidelines of the ISRU paradigm.

1. Introduction

A key challenge in creating long-term extraterrestrial habitats to support continuous human occupation lies in building structures that can bear load [1–3], while simultaneously offering protection from radiation and large temperature fluctuations [4,5]. The most abundant material for the construction of these habitats on the lunar and Martian surfaces is locally available regolith. Exploiting regolith is one part of a larger paradigm termed *in situ* resource utilization (ISRU) [6–9]. ISRU seeks to minimize dependence on terrestrial supplies and advocates the use of local resources, such as solar energy, regolith, and water, where applicable, as construction constituents [10,11]. To mimic the behavior of actual regolith, of which we have very few physical samples, several research groups around the world have developed suitable regolith simulants to reproduce compositions from different lunar or Martian sites. Some examples include Johnson Space Center (JSC) lunar regolith simulant [12], Lunar soil simulant by ISRO Sittampundi Anorthosite Complex (LSS-ISAC) [13], Lunar highland simulant (LHS), Lunar mare dust simulant (LMS), Mars global simulant (MGS) from Exolith Labs [14,15], Korea Lunar Simulant (KLS) [16], and Fuji Japan Simulant (FJS) [17].

Consolidating regolith into fundamental building blocks — akin to terrestrial bricks — requires the use of a suitable binding process and/or agents. Various consolidation methods have been explored for

extraterrestrial regolith, including thermal techniques such as conventional sintering [18,19], microwave sintering [20], and laser fusing [21], which achieve particle coalescence at elevated temperatures to produce high-strength bricks. Moreover, additive manufacturing techniques like 3D printing using photocurable lunar regolith suspensions present innovative pathways to construct complex geometries directly from regolith materials [22]. Additionally, biological approaches like Microbial Induced Calcite Precipitation (MICP) utilize urease-producing bacteria combined with organic additives such as guar gum to biologically cement regolith particles into consolidated forms with promising mechanical properties [23–26]. Emerging waterless or low-water processes, such as sulfur-based concrete formation and dry-mix steam injection methods, also offer competitive alternatives for regolith consolidation where water availability is limited [27, 28]. While these approaches have demonstrated varying degrees of success, they also present significant challenges. For example, sintering techniques require high thermal energy inputs, limiting their feasibility for resource-constrained extraterrestrial environments. The specific energy (the total net energy input required to process a unit mass of material) required with conventional sintering [29], Microwave sintering [30], and cold sintering [31] are 1.7–2.3, 15–20, and 1–3 (MJ/kg), respectively. MICP offers biogenic consolidation but involves complex biological controls and environmental sensitivities. Additive

* Corresponding author at: Department of Mechanical Engineering, Indian Institute of Science, Bangalore, 560012, Karnataka, India.

E-mail address: nitingupta@iisc.ac.in (N. Gupta).

manufacturing approaches must balance extrudability with mechanical strength. Despite these advances, a gap remains in developing scalable, low-energy, environmentally compatible methods capable of producing mechanically robust consolidates on the Moon and Mars surface.

Martian and lunar regolith share several characteristics with terrestrial soils, a fact widely recognized in current literature [32,33]. Specifically, lunar regolith can be likened to sandy soil due to its near-complete lack of inherent binding minerals (e.g., clay) as well as its poor water retention capability. On the other hand, Martian regolith resembles clayey soil more closely, possessing inherent binding materials such as hydrated silicas or gypsum [15,34]. Further pursuing the analogy with terrestrial soil, it is now known that an important class of natural binder materials, such as clay, bamboo, and related plant-based polysaccharides [35–37] can be used to bind terrestrial soil into load-bearing consolidates. Like these terrestrial binders, biopolymers represent one possible candidate among many proposed ISRU systems for extraterrestrial construction — including thermosetting/thermoplastic resins (e.g., epoxy, polyurethane, polyester, acrylic), sulfur-based binders, inorganic cements, and geopolymers — all of which must be evaluated for performance, mass, and producibility. Most such candidates currently require Earth-based synthesis and transport, as no practical in-situ process chains exist from local feedstocks, rendering them logistically expensive even with local regolith. Biopolymers stand out, however, as they could in principle be produced sustainably from biomass in controlled ecological life-support systems wherever plants are cultivated for food and oxygen. A large fraction of these natural binders are biopolymers that are also useful for sustaining plant life. Using such binders, *in lieu* of more conventional consolidation methods like kiln sintering, offers notable advantages in simplicity and energy efficiency, making it perfectly aligned with the ISRU paradigm. These biopolymer-regolith composites are thus especially attractive for early extraterrestrial construction under power-limited robotic missions, where sintering is impractical, and imported binders incur prohibitive launch-mass penalties [38]. In fact, recent work on terrestrial soils has proposed the use of biopolymers, such as guar and xanthan gums (with just 1–2 wt% of soil), as binders, resulting in compressive strengths of up to 8 MPa [39–41]. Similarly, with extraterrestrial regolith simulants, biopolymers like Bovine Serum Albumin (BSA) [42–44], and human serum albumin (HSA) [26], etc., were used to create biopolymer soil-bound composites (BSC) [45], which can reduce energy demands by orders of magnitude compared with purely thermal consolidation methods. For the construction of space habitats, the strength requirements are quite low compared with terrestrial brick. While terrestrial bricks typically exhibit compressive strengths of 20–25 MPa, design studies for extraterrestrial construction indicate that materials with compressive strengths as low as approximately 3–4 MPa for lunar and 6–7 MPa for Martian conditions may suffice for typical habitat and infrastructure designs, primarily due to reduced gravity and lower structural loading requirements [46].

In the present study, we examine the consolidation potential of regolith simulants using minimal quantities of biopolymers, leading to consolidates that we term Biopolymer Regolith Composites (BRCs). We establish a process for producing BRC bricks with consistent strength — by introducing a new method called Pressure-Assisted Die Casting (PADC), which applies pressure during the casting process to significantly enhance the mechanical performance and reduce porosity of consolidated regolith bricks, a departure from traditional casting or sintering methods. This addresses key challenges in extraterrestrial construction, including low mechanical strength and high energy consumption seen in current approaches. We position PADC as an innovative step in *in situ* resource utilization (ISRU), enabling the production of load-bearing structures on the Moon and Mars with improved durability and water resistance. Furthermore, we highlight the unique conditions of lunar and Martian regolith, such as dust cohesion and environmental extremes, and show how our method is tailored to these challenges. Unlike previous studies that mostly focus on

Table 1

Mineral composition of lunar regolith simulants.

Minerals	LHS-1	LMS-1
Anorthosite	74.4%	19.8%
Glass-rich basalt	24.7%	32.0%
Olivine	0.2%	11.1%
Pyroxene	0.3%	32.8%
Ilmenite	0.4%	4.3%
$\langle d_p \rangle$ (μm)	88	63

microbial or chemical binders, our approach leverages the mechanical advantage of pressure to form tightly bonded composites, potentially transforming local regolith into reliable building materials suitable for sustainable space habitats. In addition to the process development, we quantified the deformation and failure of BRCs using digital image correlation (DIC) techniques by tracking the crack development and detailed evolution during uniaxial compression. This enabled us to characterize displacement fields, crack initiation, and propagation with high spatial resolution, offering a more comprehensive understanding of the mechanical response beyond conventional compressive strength measures. Furthermore, we investigate the impact of varying temperature conditions on BRCs and evaluate the degradation of biopolymers under these conditions. Our objective is to establish a scalable methodology that largely utilizes *in situ* materials for construction, with a target compressive strength of 15–20 MPa, comparable to that of conventional kiln-fired bricks on Earth. This contribution advances the state of the art by providing a novel, scalable fabrication route for extraterrestrial construction materials with promising strength, durability, and sustainability characteristics.

The manuscript is organized as follows: materials used, corresponding process protocol for consolidation of BRCs, and various characterization techniques are described in Section 2. Measurement of BRC strength and its dependence on various process parameters is quantified in Section 3. Finally, we assess the sustainability of the produced bricks under simulated thermal and pressure variations. A discussion of our results with some concluding remarks is presented in Section 4.

2. Materials and methods

2.1. Regolith simulants and biopolymers

We used regolith simulants procured from Space Research Technologies (Exolith labs), USA [14,15] for our experiments. Consolidation studies were performed on four different types of simulants: 2 lunar (Lunar highland simulant — LHS-1 and Lunar mare dust simulant — LMS-1), and 2 Martian (Mars Global simulant — MGS-1 and Mars Global simulant sulfate — MGS-1S). The compositional details and mean particle size ($\langle d_p \rangle$) of these simulants are provided in Tables 1 and 2.

Three different polysaccharide biopolymers were used for binding the regolith in this study, *viz.* guar gum (GG), xanthan gum (XG), and acacia gum (AG), procured from Urban Platter, India. The selection was based on their capacity for enhancing the mechanical properties of regolith composites [23,24], compatibility with scalable and low-energy processes critical for ISRU, and potential for sustainable on-site consolidation in extraterrestrial environments. The rationale for their use is further reinforced by the possibility of biosynthetic production of analogous polysaccharides through engineered microorganisms [47], thereby supporting long-term ISRU strategies.

GG is a galactomannan polysaccharide characterized by a mannose backbone with galactose side groups, with a high molecular weight ranging from 2,00,000 to 10,00,000 [40,48]. XG is a heteropolysaccharide composed of repeating units of glucose, mannose, and glucuronic acid in a molar ratio of approximately 2:2:1, with a typical molecular weight of around 6,00,000; some fractions may exceed 10,00,000 [49].

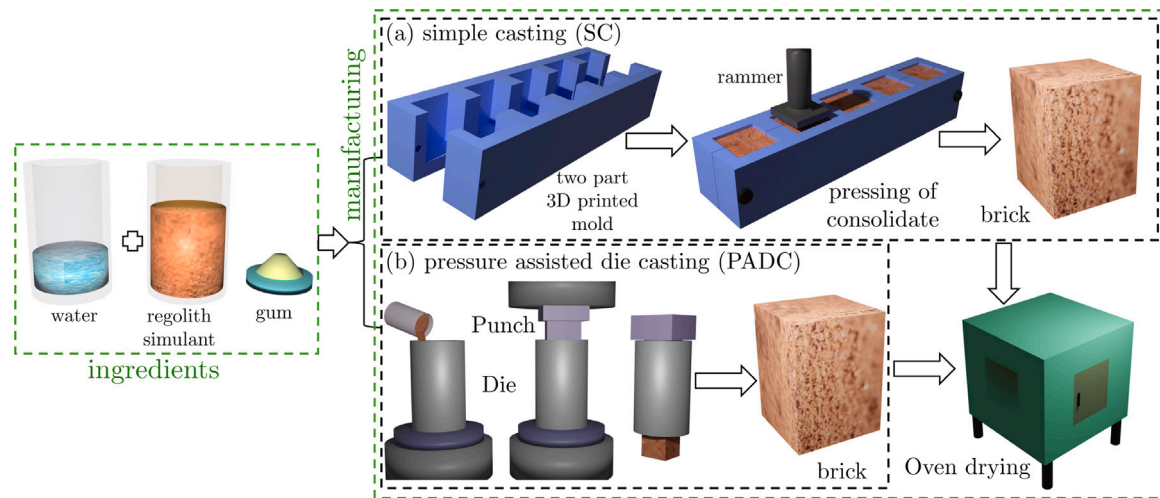


Fig. 1. Schematic showing the method for preparing biopolymer regolith composite (BRC) bricks. Row (a) shows the simple casting (SC) method wherein a two-part mold is used, and the pre-mixed material is filled and hand-rammed. Row (b) shows the pressure-assisted die casting (PADC) method for producing BRC bricks. At the end of either process, the mold/die is oven-dried at 60° for 12 h.

Table 2
Mineral composition of Martian regolith simulants.

Minerals	MGS-1	MGS-1S
Gypsum	–	40%
Plagioclase	27.1%	16.4%
Glass-rich basalt	22.9%	13.7%
Pyroxene	20.3%	12.2%
Olivine	13.7%	8.2%
Mg-sulfate	4.0%	2.4%
Ferrihydrite	3.5%	2.1%
Hydrated silica	3.0%	1.8%
$\langle d_p \rangle$ (μm)	90	119

AG primarily consists of arabinogalactan, a complex polysaccharide made up of D-galactose units with side chains of D-glucuronic acid and L-arabinose or L-rhamnose, exhibiting a molecular weight ranging from 2,00,000 to 6,00,000. GG and XG are typically produced as particles of size $\sim 180 \mu\text{m}$, with some finer grades reaching $\sim 75 \mu\text{m}$. In contrast, AG particles typically have larger sizes $\sim 200 - 500 \mu\text{m}$.

2.2. Simple casting (SC) and pressure-assisted die casting (PADC) for producing biopolymer regolith composites

Consolidation of regolith simulant using biopolymers to produce biopolymer regolith composite (BRCs) was investigated using two independent protocols, termed simple casting (SC) and pressure-assisted die casting (PADC), see Fig. 1. The starting point for both protocols was the same—a mixture of regolith simulant (100 g), polysaccharide gum powder of varying quantity (0 to 2% w/w of regolith simulant), and DI water (15 ml) was first prepared. The regolith was thoroughly mixed with the gum in powder form, after which water was added, and the resulting mixture was stirred well.

In the simple casting (SC) method, the mixture was filled into a two-part plastic mold, whose inner surfaces were covered with a transparent plastic sheet to ease sample removal. Final BRC consolidates produced from this mold had dimensions $20 \times 20 \times 20 \text{ mm}^3$, as shown in Fig. 1(a). Specimens were hand rammed and taken to oven dry at 60 °C for 12 h. Specimens were finally polished using sandpaper to obtain a flat surface for subsequent mechanical testing.

In the pressure-assisted die casting (PADC) method, the pre-fill mixture was first compacted using a hydraulic press under a load of 5 tons and die-cast into the desired shape (cubical block $15 \times 15 \times 15 \text{ mm}^3$, or cylindrical disc of diameter 12 mm with thickness 6–7 mm),

as shown in Fig. 1(b). The specimens were then dried in an oven at a temperature of 60 °C for 12 h. This post-removal drying was necessary to ensure the effective removal of moisture from the consolidated material.

The processing and usage conditions for biopolymer-regolith composites define a practical operational envelope in terms of temperature (T), pressure (p), and water availability are shown in Table 3. These parameters influence the mixing, curing, and durability of the consolidated bricks.

2.3. Mechanical testing

The strength of consolidated BRCs was quantified by measuring their unconfined compressive strength (UCS, ASTM C109) and tensile strength (Brazilian disc configuration, ASTM D3967). Both these tests were performed on a universal testing machine (UTM, Instron 5697), with 30kN (compression) and 5kN (tension) load cells under quasi-static conditions (loading rate 0.5 mm/min). The specimen dimensions were smaller than ASTM standards because of practical constraints, but satisfied the plane stress condition for compression and tensile testing.

The compressive strength was calculated as $\sigma_c = \frac{P_{c,max}}{A_c}$, where $P_{c,max}$ is the maximum load at failure, and A_c is the initial cross-sectional area of the specimen. The tensile strength was calculated as $\sigma_t = \frac{2P_{t,max}}{\pi D t}$, where $P_{t,max}$ is the maximum load at failure, and D and t are the initial diameter and the thickness of the specimen, respectively [50]. Both measurements were averaged over a minimum of four specimens, with standard deviation reported in the corresponding error bars, and each mechanical test was repeated three times independently to ensure repeatability.

To quantify failure under loading, crack paths were identified, and corresponding strains were evaluated using digital image correlation (DIC) analysis. For this purpose, digital images were recorded using a DSLR camera (Nikon D850) at 25 FPS and analyzed frame-by-frame using the Ncorr software package [51] with MATLAB version 2021a. The need to ensure planar deformation and through-thickness cracks (required for DIC) meant that these analyses could only be performed for the tensile samples. The cubical blocks used for compression tests inevitably failed because of multiple cracks developing through the thickness and out of the viewing plane of the camera.

Table 3
Operational envelope for manufacturing BRCs.

Parameter	Feasible Range	Comments
Temperature (T)	Ambient to 60 °C during curing	Oven drying at 60 °C for 12 h; bricks maintain strength under temperature cycling from −50 to +70 °C typical of extraterrestrial surfaces.
Pressure (P)	Atmospheric pressure during processing; operation feasible under lunar/Martian low pressures once consolidated	Consolidation via hand-ramming or PADC at Earth pressure; bricks stable under vacuum and low-pressure conditions of Moon/Mars after curing.
Water Availability	Approximately 15 ml water per 100 g regolith (15% w/w ratio) needed for mixing	Water is essential for binder hydration and mixing; water recovery and recycling necessary in closed loop extraterrestrial missions.

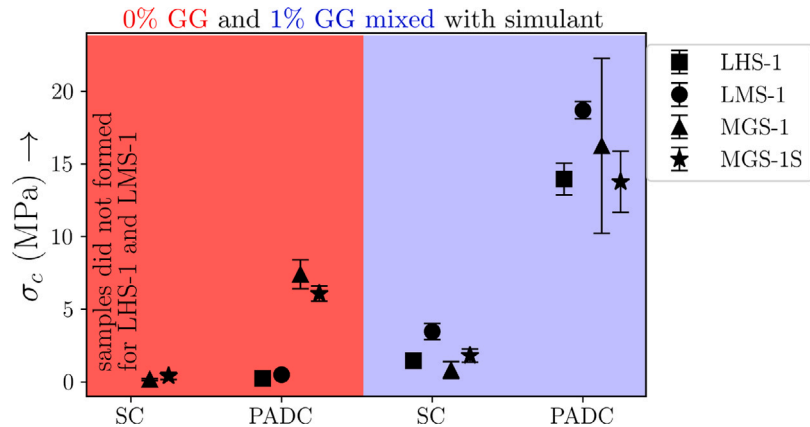


Fig. 2. Compressive strengths σ_c of regolith simulants with the SC/PADC protocols as well as with (red) and without (blue) biopolymer binders (1% w/w guar gum). The average strength was obtained from four sample measurements, with error bars representing one standard deviation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Results

We present the results of various process and consolidation parameters on the final mechanical properties of biopolymer regolith composites (BRCs). The role of different biopolymer binders in determining BRC strength is also quantified, and failure mechanisms under load application are evaluated.

3.1. Strength of simple (SC) and pressure-assisted die cast (PADC) bricks

Compressive strengths (σ_c) of BRC bricks produced via simple casting (SC) and pressure-assisted die casting (PADC) of lunar simulants LHS-1, LMS-1, and Martian simulants MGS-1 and MGS-1S are reproduced in Fig. 2. In the left half (red) of the plot, samples were cast using only water and no biopolymer binder, while the right half (purple) is samples with 1% w/w of guar gum (GG). The same markers are used for LHS-1 ($\square$), LMS-1 ($\circ$), MGS-1 ($\triangle$) and MGS-1S ($\star$) in both halves. When no binders are used, it is clear that samples produced via SC were incapable of bearing any load. In fact, it was not even possible to obtain consolidated samples with LHS-1 and LMS-1. On the other hand, weakly consolidated samples were obtained from MGS-1 and MGS-1S even in the absence of biopolymer binders, owing to the presence of inherent, but weak, binders in the simulant itself (1.8%–3% hydrated silica). With the PADC process, a significant increase was observed in the compressive strengths of all four regolith simulants, with the largest seen for the Martian simulant samples. The hydrated silica (C-H-S) in these simulants effectively binds loose soil particles under high-pressure and moisture conditions [52,53]. However, despite this increase in compressive strengths of over 5 MPa, the resulting samples were water-soluble and would degrade within minutes when put in water.

When combined with 1% w/w of guar gum, both the compressive strength and final water resistance were significantly enhanced. All four

simulants could be consolidated with the SC protocol, but showed compressive strengths of not more than a few MPa. On the other hand, the PADC process produced bricks with $\sigma_c \sim 15$ MPa. This is to be expected since it is known, for instance, that biopolymers, especially polysaccharides, can form stable aggregates with terrestrial soils [54,55]. However, the mechanisms and effectiveness of the binding are different when comparing soils with and without inherent binders [36,39–41]. In the former case (as with MGS-1 and MGS-1S), polysaccharides predominantly adhere via weak hydrogen bonds between the carboxyl group and the hydroxyl group of the polysaccharide molecules and cations in the soil. This interaction facilitates the formation of bridges between binding mineral particles, forming microaggregates (size < 250 microns) and macroaggregates (size > 250 microns). Conversely, when no inherent binders are present (as with LHS-1 and LMS-1), polysaccharides create a cross-linked three-dimensional network, resulting in inter-particle cohesion. Having established the importance of the polymer binder (GG), we now investigate the large increase in strength of bricks produced via PADC vis-à-vis the SC process. This can be attributed to the reduction in part porosity under the application of pressure. Optical micrograph images of samples produced via SC and PADC (1% GG) are reproduced in Fig. 3 for LHS-1 simulant. While the overall porosity (dark) is evidently lesser in the PADC-produced part (panel b), the inset to these two panels shows the difference more dramatically. It is well-known that the compressive strength of concrete-like materials primarily depends on the porosity of the structure. In this case, let us consider a scenario where the overall porosity of bricks is nominally 25% [18] under PADC (measured using mercury-intrusion porosimetry) after compaction under a hydraulic press. Using the exponential relationship between porosity and strength given in Ref. [19], we estimate that the porosity in the SC case is nearly 50%.

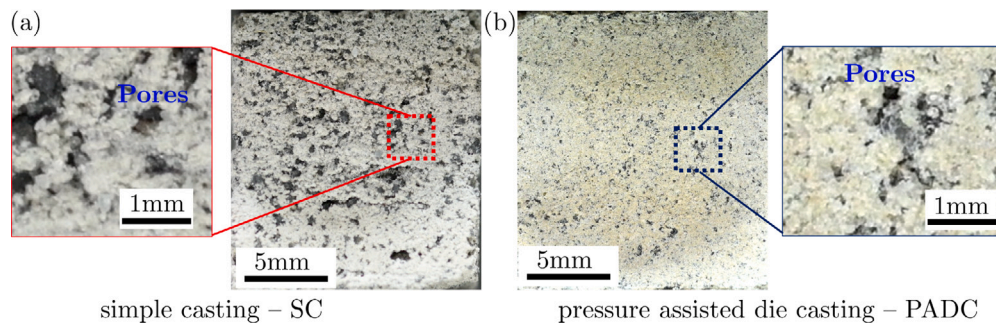


Fig. 3. Optical image showing LHS-1 based consolidated BRC brick, with 1% w/w guar gum. (a) The sample was obtained via simple casting (SC) and (b) pressure-assisted die casting (PADC). The inset to each image clearly shows the lowered porosity in the PADC case due to the applied compaction pressure.

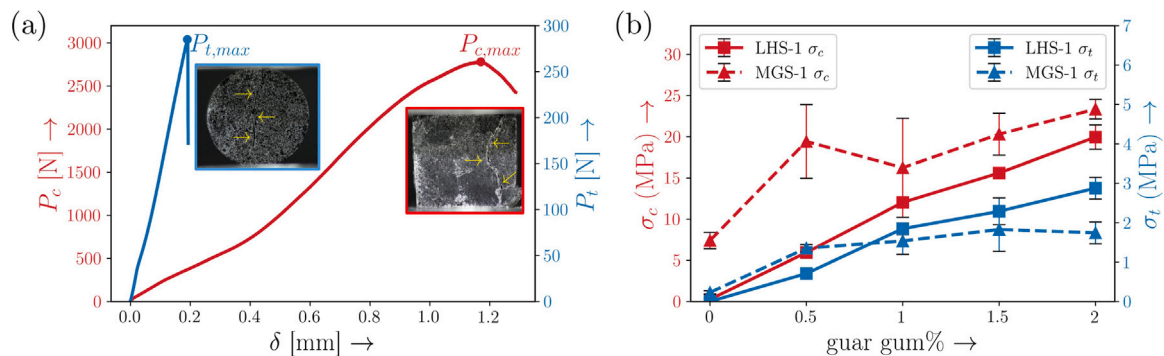


Fig. 4. Compressive and tensile strengths of BRC bricks and their variation with binder concentration. (a) Typical load–displacement curves are used to determine the compressive and tensile strengths of BRC bricks. The inset shows a crack pattern at the point of failure, corresponding to the maximum load in tension or compression. (b) Variation in compressive (σ_c) and tensile (σ_t) strengths as a function of guar gum concentration. Data is for specimens (LHS-1 or MGS-1) made using the PADC process (5-ton load).

3.2. Does the quantity of gum matter?

The second question that we attempt to answer pertains to the optimal concentration of biopolymer in the final BRC. For this, guar gum (GG) content varied systematically, and corresponding compressive and tensile tests were performed. Since both lunar simulants (LMS-1 and LHS-1) and Martian simulants (MGS-1S, MGS-1) demonstrated similar behavior among themselves, we only report results for LHS-1 and MGS-1 henceforth, unless specified otherwise. Further, only results with the PADC process are reported since the strengths obtained are superior to those from the SC process.

Typical load–displacement graphs for compression (P_c , red) and tension (P_t , blue) are shown in Fig. 4(a). The compressive (tensile) strength σ_c (σ_t) was obtained from the corresponding peak $P_{c,max}$ ($P_{t,max}$). Insets to this figure show the occurrence of cracks at the maximum load; it is clear from them that the tensile specimens (blue) failed via a single through-thickness crack (yellow arrows), while compressive specimens (red) failed via multiple crack propagation through the sample thickness (yellow arrows).

Variation in σ_c (red) and σ_t (blue) with GG concentration is shown in Fig. 4(b). It is clear at once that increased GG concentration leads to higher tensile and compressive strengths for both LHS-1 and MGS-1. Initially, a nearly two-fold increase in σ_c with GG content from 0.5% to 1% by weight was observed with LHS-1 ($\square$); similar trends were also observed for σ_t , see solid lines in Fig. 4(b). In contrast, MGS-1 ($\triangle$) showed a saturation in both σ_c and σ_t with GG content around 1%, see dashed lines in Fig. 4(b). The optimal GG content was hence determined to be around 2% and 0.5% for lunar and Martian regolith, respectively.

It is perhaps likely that this difference in σ_c, σ_t dependence on gum concentration is due to the nature of binding alluded to earlier. It is also important to note that for concentrations above 2%, material compaction during the PADC became significantly challenging. This

precluded testing for concentrations above 2%. Water content in the pre-fill mixture is another important parameter—insufficient water can impede proper hydration and bonding, while excessive water can weaken the structure by increasing voids.

3.3. Failure mechanisms and crack propagation in BRCs

As discussed in Section 2, we used digital image correlation (DIC) to analyze failure in the tensile test samples. This is not done for the compressive samples because the sample thickness mandated by the test standard results in out-of-plane deformation. The results for three different gum concentrations (GG from 0.5% to 1.5%) are summarized in Fig. 5 and pertain to the LHS-1-based BRC brick. The conclusions to be presented were found to be very similar for the other simulants as well.

In panel (a) of this figure, three different loads (P_t) vs. displacement (δ) graphs are presented, with varying GG concentrations as indicated in the legend. Specifically, each curve has three points marked—the first during deformation and well before failure (A, P, X), the second very close to the point of failure (B, Q, Y), and the third post-failure and load reduction (C, R, Z). Corresponding images of the brick, with tensile strain superimposed, are shown in panel (b). Here, the tensile strain component (ϵ_{xx} , say) computed is along an axis (x) perpendicular to the loading direction (y). As a result, cracks correspond to large values of the ‘opening strain’ ϵ_{xx} . Additionally, the strain fields are computed by comparing the current image with a reference image before deformation starts, and they correspond to the total (and not incremental) Lagrangian strain.

It is clear from the results for all three concentrations that failure inevitably occurs via the nucleation of a single crack at the loading platens (see images corresponding to points B, Q, Y) that then

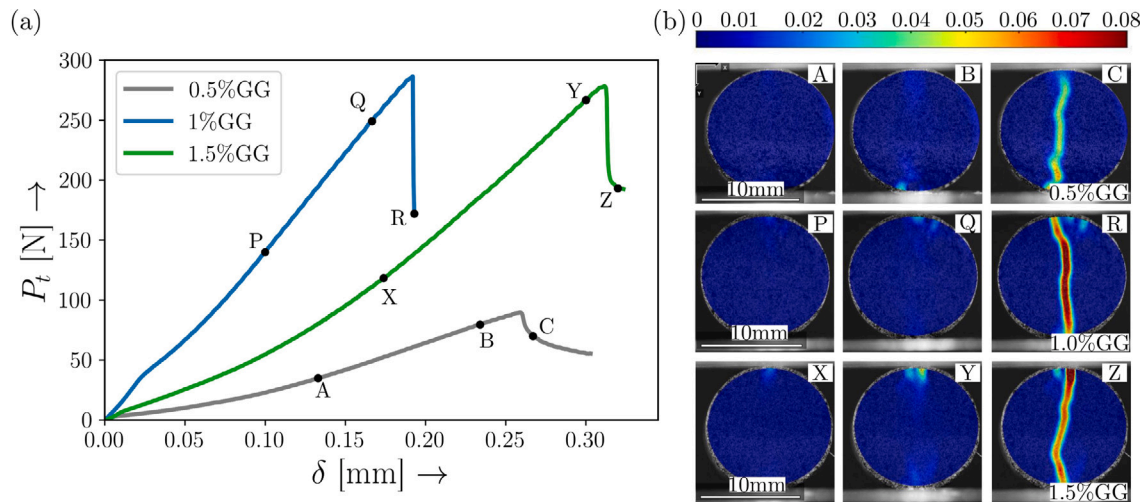


Fig. 5. Failure of BRC bricks made from LHS-1 under tensile loading. (a) Load displacement graphs for three different guar gum concentrations, each with three points, corresponding to snapshots well before maximum load (A, P, X), close to maximum load (B, Y, Q), and just following failure (C, R, Z). The corresponding snapshots, with superimposed tensile strain fields (from DIC), are reproduced in panel (b). Color represents the Lagrangian strain transverse to the loading direction. Note that the failure pattern is unchanged by the binder concentration, and only the strain at failure is altered. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

propagates largely linearly in the direction of loading, see images corresponding to points C, R, Z in panel (b). By comparing the data in panels (a) and (b) of Fig. 5, it is clear that the mode of failure is independent of the gum/ binder concentration and occurs via a single crack along the loading direction. The tensile and compressive strengths generally improve with biopolymer content, but strain at failure may exhibit non-monotonic trends due to processing and material heterogeneity effects. The similarity in strength observed between 1% and 1.5% guar gum, given experimental variability, aligns with the literature reporting an optimal binder concentration for maximal mechanical performance [37,56,57].

3.4. What about other bio-polymer binders?

To compare the dependence of BRC strength on binder chemistry, we evaluated bricks produced via PADC with three different biopolymer binders — guar gum (GG), xanthan gum (XG), and acacia gum (AG) — at a fixed concentration of 1% w/w of the simulant (0.4 g gum powder in 40 g of simulant). Results of compressive σ_c and tensile σ_t strength evaluation of these samples are shown in Fig. 6. The error bars in this plot represent the standard deviation from four sample measurements.

Upon comparing the strengths of the four different regolith simulants, it is clear that both GG and XG perform comparably with all simulants, and much better than AG. This is notable since GG and XG have been identified as being excellent binders owing to their high molecular weights, hydrophilicity, and stability under varying conditions [40,49]. When mixed with water, XG forms a stable hydrophilic viscous colloid, while GG demonstrates thickening and gelling properties upon dissolution. Although AG is soluble in water, its binding and thickening capabilities are generally inferior to those of GG and XG. This enhancement is attributed to the coating and binding actions of polysaccharides on sand particles, which contribute to improved moisture retention within the regolith.

Finally, the dashed line in Fig. 6 represents $\sigma_t = 0.1\sigma_c$; it is seen that the strength values for all three polysaccharide gums appear to fall close to this line. This relationship is consistent with prior studies on terrestrial soils [58]; for instance, in concrete, the tensile strength can range from approximately 7% to 15% of the compressive strength. This characteristic is essential because while concrete performs well under compressive loads, it is relatively weak in tension, notably leading to failure under bending loads.

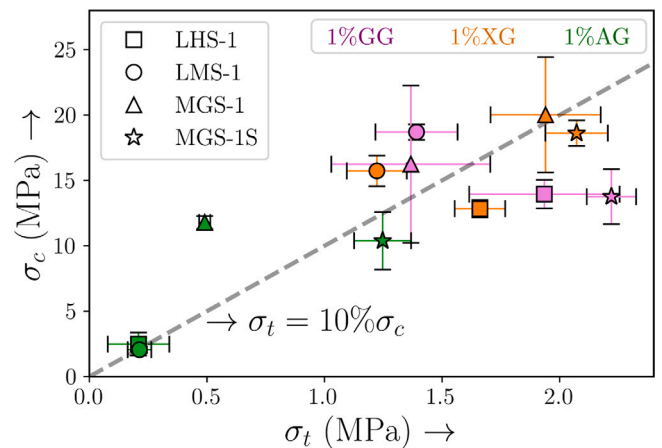


Fig. 6. Scatter plot showing compressive (σ_c) and tensile (σ_t) strengths for different simulants ($\square$ — LHS-1, $\circ$ — LMS-1, $\triangle$ — MGS-1, and $\star$ — MGS-1S) mixed with guar gum (violet), xanthan gum (orange) and acacia gum (green), respectively. The marker shows the average strength (tested for four specimens), and the error bar represents the standard deviation. All samples were made using PADC (5 tons compaction), followed by 60 °C oven drying for 12 h. The dashed black line represents the relationship $\sigma_t = 0.1\sigma_c$, and mostly lies in 0.07 to 0.15 σ_t . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.5. Integrity of brick under simulated environment

For BRCs to be eventually usable in extraterrestrial environments, they must be able to endure large temperature fluctuations as well as low pressures. This endurance can be evaluated in two different ways—first, the effect of such environmental fluctuations on the pre-fill mixture itself, which determines the possibility of the process being used under these conditions. The second is to evaluate the resilience of the finally produced BRC bricks themselves to these fluctuations. We comment on both of these modalities in this section.

3.5.1. Role of temperature fluctuations

Temperatures on the lunar and Martian surfaces range from -223 °C to 123 °C, and from -140 °C and 20 °C, respectively. It is clear that

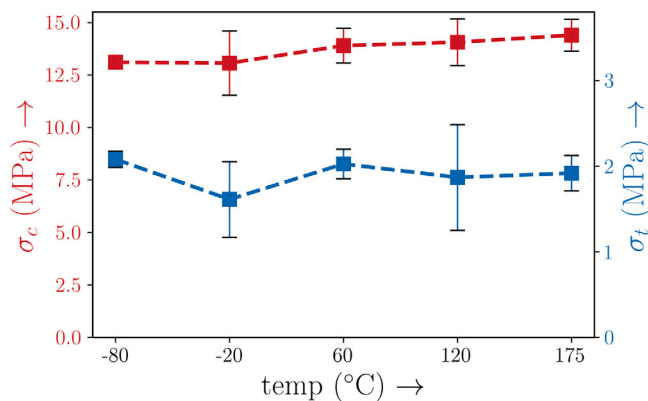


Fig. 7. Variation in compressive σ_c and tensile σ_t strengths of LHS-1 based BRC bricks (1% w/w guar gum, PADC process) with temperature. Brick strength, post-casting, is seen to be largely unchanged by large temperature variations.

producing BRC consolidated bricks is not possible for this entire range; for instance, guar gum by itself can disintegrate beyond 80 °C. Hence, we evaluate the effect of large and small temperatures on already consolidated BRC bricks. Results are reported for LHS-1 with 1% GG using the PADC process; analogous results were obtained with the other simulants as well.

We subjected LHS-1 based bricks to a temperature of 175 °C in a hot air oven for 12 h, as well as -80 °C in a deep freezer for 12 h. This time duration was chosen to be concomitant with the time required for brick formation. Additional intermediate temperatures were also chosen—we selected soaking temperatures of -80 °C, -20 °C, 60 °C, 120 °C, and 175 °C. Following this soaking, the samples were subjected to mechanical testing to evaluate any changes in σ_c , σ_t , and their corresponding results are summarized in Fig. 7. It is clear from this figure that the bricks continued to retain their room-temperature strength even after being subjected to these large temperature changes; both σ_c and σ_t did not change by more than a few percent, well within experimental error. It is therefore safe to conclude that, while the bricks have to be cast at room temperature to prevent degradation of the biopolymer constituents, once formed, they are capable of withstanding large temperature fluctuations with no measurable change in their strength. It is likely that the strong bonds and low porosity resulting from the PADC process are responsible for this robust strength.

3.5.2. Can we produce BRC in vacuum?

To now evaluate if the bricks can themselves be produced under vacuum environments, we subjected SC consolidated bricks, prior to hardening, to a vacuum environment, see Fig. 8(a). Given the varied chemistry of the simulants, including their water retention ability and presence of inherent binders, separate samples were prepared using LHS-1, LMS-1, and MGS-1 simulants, each combined with 0% and 1% guar gum (GG) and DI water, see Figs. 8(b) and (c), respectively. The pressure for consolidation was set to be ~ 20 mbar (same as the pressure on the surface of Mars), in contrast with the consolidation performed at room temperature (1 bar), as discussed in Section 3.1. At this low pressure, the boiling point of water is significantly reduced to ~ 15 °C, leading to water evaporation from within the pre-fill mixture. As a result, the samples completely collapsed in the absence of any binder, see corresponding images in panel (b). It is clear that the Martian simulant, with its inherent binder, was somewhat immune to this effect.

On the other hand, the presence of GG binder appeared to minimize this effect, resulting in BRC consolidates that did not simply disintegrate, see panel (c). However, additional pores did form where water escaped the surface (see arrows in each image). It is interesting that these marks were particularly pronounced with LMS-1 when compared

Table 4

The compressive strength (σ_c) of LHS-1, LMS-1, and MGS-1 consolidates mixed with 1% GG, exposed to vacuum conditions (soaking pressure ranging from 5 to 20 mbar), compared to bricks tested under ambient pressure (1 bar).

Simulant	Soaking pressure (mbar)	σ_c (MPa)
LHS-1	1000	10.7 ± 0.7
	5–20	10.1 ± 0.8
LMS-1	1000	19.9 ± 1.8
	5–20	16.7 ± 0.4
MGS-1	1000	17.9 ± 1.2
	5–20	15.4 ± 5.1

with both MGS-1 and LHS-1. It thus appears that bricks can be produced even via the SC process under vacuum conditions. The possibility of using PADC for minimizing water loss under vacuum conditions is one that we could not evaluate in the present study. However, based on the earlier results, we believe that this should enable brick production under these extreme environments.

To evaluate the performance of already formed BRC bricks (1% GG, PADC) under vacuum conditions, all three bricks — LHS-1, LMS-1, and MGS-1 — were subjected to a pressure ~ 20 mbar for 24 h. Visual inspection of the bricks before and after this exposure revealed no observable changes in the surface porosity or structure, see corresponding images in Fig. 9 for LHS-1 in particular. Following this soaking, the compressive strength of the bricks was again evaluated, and the results are reproduced in Table 4, in comparison with room pressure testing (1 bar). It is clear from these results that the bricks do not undergo any observable strength degradation after being subjected to vacuum conditions. The use of 1% GG binders hence appears to result in bricks that can withstand both temperature and pressure fluctuations as long as they are produced within a controlled environment.

4. Discussion & summary

Our study shows that polysaccharide biopolymers can effectively consolidate lunar and Martian regolith simulant (leveraging locally available resources) into consolidated load-bearing structures, aligning with the in situ resource utilization (ISRU) paradigm and reducing reliance on Earth-derived materials. Specifically, the use of pressure-assisted dice casting (PADC) route can result in bricks — termed biopolymer regolith composite (BRC) bricks — with compressive strengths in excess of 15 MPa from both lunar and Martian simulants (surpassing the minimum structural requirements (3–7 MPa) for lunar and Martian habitats under reduced gravity), and tensile strengths near 2 MPa, maintaining mechanical integrity under extraterrestrial environmental conditions. The possible applications based on the strengths are shown in Table 5. Under conditions where (i) only modest structural performance is required (e.g., non-pressurized shielding elements, trafficable pavers, small berms), (ii) power availability is limited, and (iii) it is feasible to import or locally produce small quantities of polysaccharide gums, biopolymer consolidation becomes an attractive ISRU option, trading peak strength for dramatically reduced energy and equipment demands.

PADC has minimal energy requirements when compared to processes such as furnace (or kiln) sintering, and is therefore much more easily deployed in remote and/or extraterrestrial environments. Unlike other methods, PADC does not demand precise control of processing parameters. However, its compressive strength remains lower than that of some ISRU materials like geopolymer concretes, which can exceed 20 MPa. The resilience of BRCs to ultra-high vacuum and extreme thermal cycling, combined with their high compressive/tensile strengths at minimal binder content, positions biopolymers as the frontrunner for scalable extraterrestrial construction—especially when modest structural demands (e.g., landing pads, berms) align with constrained power

Table 5
Mechanical strength required for possible applications on the extraterrestrial surface.

Strength level	Example Methods	Suitable applications	Limitations	Refs
5–10 MPa	MICP, low-end BRC	Shields, non-load partitions, road base	Avoid spans >1 m or heavy equipment	[24]
10–20 MPa	Optimized BRC	Walls, floors, habitat infill; most early needs	Not for primary beams/columns	This work
30–50 MPa	Microwave sinter, fast-firing	Load-bearing walls, short spans	Higher energy; overkill for basics	[18,30]
100+ MPa	Full kiln melt, laser sinter	High-load beams, seismic zones	Power/equipment heavy	[59]

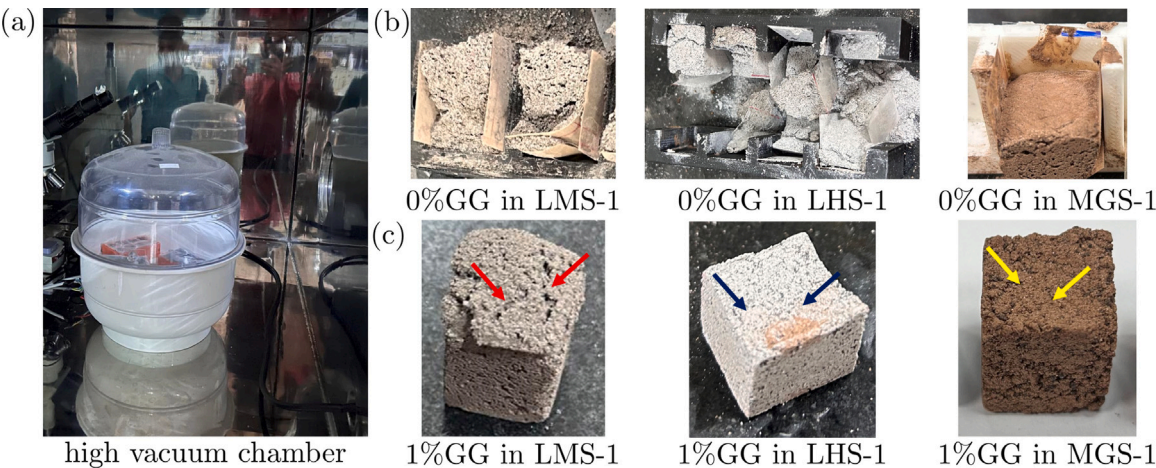


Fig. 8. Effect of vacuum conditions on BRC brick production and use. (a) Simple cast (SC) samples from LHS-1, LMS-1, and MGS-1 are placed inside a desiccator and then placed within a vacuum chamber. Panels (b) and (c) show the SC specimens under low pressure (~20 mbar) for LHS-1, LMS-1, and MGS-1, with 0% and 1% binder concentration, respectively.

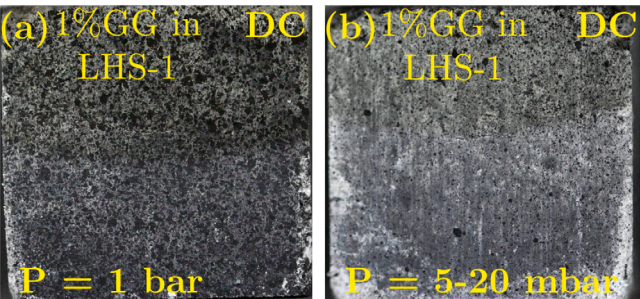


Fig. 9. Comparison of the surface texture of LHS-1 PADC specimen, using the PADC process. (a) Surface prior to vacuum exposure and (b) after being placed in a vacuum for 24 h.

budgets of <1 kW. Unlike sintering (energy-intensive, prone to microcracking) or MICP (microbe viability issues), biopolymers require no live cultures, ovens, or furnace, reducing imported mass by >90% and enabling robotic deployment with off-the-shelf compactors. This economic edge — coupled with proven 10–15 MPa performance — makes BRCs not just viable, but optimal for sustainable lunar/Martian bases under realistic early-mission limits. Furthermore, the production of biopolymers on the Moon or Mars depends on advanced biomanufacturing infrastructure and controlled agriculture, limiting near-term applicability.

PADC process is inherently more scalable and, therefore, in line with the *in situ* resource utilization (ISRU) paradigm. On scaling up the specimen dimensions (from the laboratory to practical applications), mechanical properties may be affected due to factors including

increased probability of pores and inhomogeneities becoming influential at larger scales. Larger specimens could exhibit reduced strength compared to smaller test samples as stress concentrations and failure mechanisms become more complex. However, the PADC process supports scalability by ensuring consistent densification, consistent strength and homogeneous binder distribution, which can mitigate some scale-related reductions in mechanical performance. Bricks retain strength under extreme temperature variations and low pressures, ensuring suitability for space environments. Various polysaccharide biopolymers, such as guar and xanthan gum, successfully consolidate regolith simulants, offering flexibility in binder selection.

The primary motivation for choosing biopolymers as the basic binding material is that, unlike conventional cement and synthetic polymer binders that have been extensively studied for lunar and Martian regolith stabilization, biopolymers represent a sustainable, low-energy alternative that could be locally sourced or biologically produced in space habitats. Recent work [60] indicates that certain biopolymers can be derived from terrestrial plants, seaweed, or microbial processes that might be replicable and feasible to replicate on the lunar or Martian surface, significantly reducing dependence on Earth-supplied materials. In addition to this, biopolymers also offer advantageous mechanical and adhesive properties, resilience in extreme extraterrestrial conditions [61,62], and compatibility with additive manufacturing methods such as extrusion-based 3D printing, which are critical for automated habitat construction under crew-limited scenarios.

The performance of lunar and martian simulants with the PADC process utilizing biopolymer binders is somewhat different due to their inherently varied chemistry. In the case of Martian simulants, optimal strength is reached for guar gum concentrations of around 0.5%, whereas in lunar simulants, this value is close to 2%. The application of pressure in the PADC process precludes the use of higher binder

Table 6
Comparison of regolith compaction techniques for lunar and martian construction.

Parameter	Biopolymer- Regolith Composite	Sintering	Geopolymer Concrete	Laser Melting/3D Printing	Extrusion Based AM (Concrete-like)
Fabrication Speed	Moderate; batch pressing hours	Slow; hours to days for sintering	Moderate; hours to days curing	Slow to ; moderate layer-by-layer	Moderate to fast; continuous extrusion
Building Precision	Moderate; limited by die and shape	High; thermally controlled	Moderate; depending on mold/print setup	High; Precise laser control	Moderate; nozzle limits resolution
Energy Consumption	Low; ambient/low temperature curing	High; requires high temperature	Moderate; moderate curing energy	High; laser power- intensive	Moderate; electrical power for extrusion
Regolith Infill	90% regolith simulant	~ 100% regolith	75% regolith + binder	100% regolith powder	85%–90% regolith
Water Use	Required for binder mixing and curing	None required	Required for geopolymer synthesis	None required	Required for slurry
Mechanical Strength	~15 MPa compressive; 2 MPa tensile	~ 230 MPa compressive	15–40 MPa compressive	50–70 MPa compressive	15–30 MPa compressive
Advantages	Low energy, sustainable, resilient to temp cycling	Very strong, dense structures	Durable, established cement alternatives	High precision, strong, complex shapes	Automated, scalable, robust
Disadvantages	Low strength	Energy intensive, slower process	Requires chemical additives and curing	High energy, complex equipment	Moderate energy, binder logistics
References	This work	[63]	[64]	[65]	[66,67]

concentrations with both types of simulants. It is expected that the presence of inherent binders in Martian simulants results in slightly larger compressive and tensile strengths compared to lunar simulants. Furthermore, a change in binder concentration does not fundamentally change the failure mechanisms in BRC bricks, but only the strain (or, consequently, stress) at failure. This was confirmed via simultaneous force measurements and DIC analysis. Table 6 provides context for selecting construction methods by balancing mechanical performance, operational feasibility, energy constraints, and resource availability for lunar and Martian settlement scenarios.

Notably, this study introduces the novel PADC method, which applies pressure during casting to enhance composite density and mechanical performance, distinguishing it from previous consolidation techniques that lack pressing steps. This innovation addresses challenges such as controlling porosity and mechanical heterogeneity in extraterrestrial regolith materials. The incorporation of biologically derived polymers like guar gum, combined with microbial induced calcite precipitation, further elevates mechanical strength and environmental stability, reflecting a bioinspired fabrication strategy suited for scalable habitat construction on the Moon and Mars. Other polysaccharide biopolymers, including xanthan gum, also effectively bind regolith in this process. The sustainable and low-energy nature of the PADC process and biopolymer binders offers advantages over energy-intensive sintering or melting methods, aligning with the stringent resource constraints of lunar and Martian environments. Despite some previously studied ISRU materials demonstrating higher absolute strengths, many rely on processes less practical for space applications. Our approach balances compressive strength, durability, environmental resilience, and operational feasibility under harsh extraterrestrial conditions, making BRC bricks promising candidates for infrastructure development. While the process requires some temperature and pressure control (to prevent binder degradation), the final bricks themselves can be subjected to significant temperature and pressure variations, with little to no change in their compressive or tensile strengths. We believe that this makes BRC-based bricks a suitable candidate for use in harsh

extraterrestrial environments while adhering to the guidelines of the ISRU paradigm.

These findings hold important implications for the development of sustainable infrastructure on the Moon and Mars, highlighting the potential to leverage local resources and biological materials to reduce dependency on Earth-supplied construction inputs while ensuring structural durability in harsh lunar and Martian environments. This work also utilized DIC in elucidating microstructural evolution under load and complementing quantitative metrics, thereby reinforcing the significance of PADC-manufactured regolith bricks. Future work should explore the long-term durability and performance of biopolymer-regolith composites under dynamic space conditions and extend the fabrication techniques toward additive manufacturing processes for in situ habitat assembly.

CRediT authorship contribution statement

Nitin Gupta: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aloke Kumar:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Koushik Viswanathan:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge financial support (AK for DST/TDT/AM/2022/345(G)), and (KV for DST/TDT/AM/2022/280(G)) from the Department of Science and Technology–DST, Govt. of India, under the Advanced Manufacturing Technologies program, for carrying out this work. NG gratefully acknowledges the support and insightful discussions provided by the LAMFIP Research Group of the Department of Mechanical Engineering at IISc Bangalore.

References

- [1] N. Barlow, MARS: An Introduction to Its Interior, Surface and Atmosphere, USA, 2008.
- [2] C.S. Cockell, M. Balme, J.C. Bridges, A. Davila, S.P. Schwenzer, Uninhabited habitats on Mars, *Icarus* 217 (1) (2012) 184–193.
- [3] L.T. Elkins-Tanton, The Earth and the Moon, Infobase Publishing, 2006.
- [4] C. Leovy, Weather and climate on Mars, *Nature* 412 (6843) (2001) 245–249.
- [5] F. Gifford Jr., The surface-temperature climate of Mars., *Astrophys. J. Vol.* 123, P. 154 123 (1956) 154.
- [6] J.-M. Gosau, Regolith stabilization and building materials for the lunar surface, in: *Earth and Space 2012: Engineering, Science, Construction, and Operations in Challenging Environments*, 2012, pp. 243–249.
- [7] Y.C. Toklu, P. Akpinar, Lunar soils, simulants and lunar construction materials: An overview, *Adv. Space Res.* 70 (3) (2022) 762–779.
- [8] M. Naser, Extraterrestrial construction materials, *Prog. Mater. Sci.* 105 (2019) 100577.
- [9] I.A. Crawford, K.H. Joy, Lunar exploration: opening a window into the history and evolution of the inner solar system, *Philos. Trans. R. Soc. A: Math. Phys. Eng. Sci.* 372 (2024) (2014) 20130315.
- [10] D. Biswas, Efficient Sintering of Lunar Soil using Concentrated Sunlight, The University of Maine, 2023.
- [11] H. Benaroya, Turning Dust to Gold: Building a Future on the Moon and Mars, Springer Science & Business Media, 2016.
- [12] D.S. McKay, J.L. Carter, W.W. Boles, C.C. Allen, J.H. Allton, JSC-1: A new lunar soil simulant, *Eng. Constr. Oper. Space IV* 2 (1994) 857–866.
- [13] I. Venugopal, K. Muthukumar, K. Sriram, S. Anbazhagan, T. Prabu, S. Arivazhagan, et al., Invention of Indian moon soil (lunar highland soil simulant) for chandrayaan missions, *Int. J. Geosynth. Ground Eng.* 6 (2020) 1–9.
- [14] M. Isachenkov, S. Chugunov, Z. Landsman, I. Akhatov, A. Metke, A. Tikhonov, et al., Characterization of novel lunar highland and mare simulants for ISRU research applications, *Icarus* 376 (2022) 114873.
- [15] K.M. Cannon, D.T. Britt, T.M. Smith, R.F. Fritsche, D. Batchelder, Mars global simulant MGS-1: A rocknest-based open standard for basaltic martian regolith simulants, *Icarus* 317 (2019) 470–478.
- [16] B.-H. Ryu, C.-C. Wang, I. Chang, Development and geotechnical engineering properties of KLS-1 lunar simulant, *J. Aerosp. Eng.* 31 (1) (2018) 04017083.
- [17] H. Kanamori, S. Udagawa, T. Yoshida, S. Matsumoto, K. Takagi, Properties of lunar soil simulant manufactured in Japan, *Space* 98 (1998) 462–468.
- [18] N. Gupta, V. Dawara, A. Kumar, K. Viswanathan, Synthetic space bricks from lunar and martian regolith via sintering, *Adv. Space Res.* (2024).
- [19] N. Gupta, V. Dawara, A. Kumar, K. Viswanathan, Liquid state sintering enhances consolidation in basalt-rich lunar regolith, *Adv. Space Res.* (2024).
- [20] Y.-J. Kim, B.H. Ryu, H. Jin, J. Lee, H.-S. Shin, Microstructural, mechanical, and thermal properties of microwave-sintered KLS-1 lunar regolith simulant, *Ceram. Int.* 47 (19) (2021) 26891–26897.
- [21] K.W. Farries, P. Visintin, S.T. Smith, Direct laser sintering for lunar dust control: An experimental study of the effect of simulant mineralogy and process parameters on product strength and scalability, *Constr. Build. Mater.* 354 (2022) 129191.
- [22] M. Isachenkov, S. Chugunov, I. Akhatov, I. Shishkovsky, Regolith-based additive manufacturing for sustainable development of lunar infrastructure—An overview, *Acta Astronaut.* 180 (2021) 650–678.
- [23] R. Dikshit, A. Dey, N. Gupta, S.C. Varma, I. Venugopal, K. Viswanathan, et al., Space bricks: From LSS to machinable structures via MICP, *Ceram. Int.* 47 (10) (2021) 14892–14898.
- [24] R. Dikshit, N. Gupta, A. Dey, K. Viswanathan, A. Kumar, Microbial induced calcite precipitation can consolidate martian and lunar regolith simulants, *PLoS One* 17 (4) (2022) e0266415.
- [25] N. Gupta, R. Kulkarni, A.R. Naik, K. Viswanathan, A. Kumar, Bacterial bio-cementation can repair space bricks, *Front. Space Technol.* 6 (2025) 1550526.
- [26] A.D. Roberts, D. Whittall, R. Breitling, E. Takano, J.J. Blaker, S. Hay, et al., Blood, sweat, and tears: extraterrestrial regolith biocomposites with in vivo binders, *Mater. Today Bio* 12 (2021) 100136.
- [27] M. Fateri, A. Gebhardt, Process parameters development of selective laser melting of lunar regolith for on-site manufacturing applications, *Int. J. Appl. Ceram. Technol.* 12 (1) (2015) 46–52.
- [28] K. Snehal, P. Sinha, P. Chaunsali, Development of waterless extra-terrestrial concrete using Martian regolith, *Adv. Space Res.* 73 (1) (2024) 933–944.
- [29] J. Bowen, V.L. Prabhu, S. Lim, M. Anand, The viscosity and processing of molten lunar regolith, *Sci. Rep.* 15 (1) (2025) 3938.
- [30] Z. Liu, J. Li, X. Yang, Y. Ding, S. Meng, Sintering of lunar regolith simulants using a domestic microwave: A rapid and energy-efficient strategy for construction material fabrication, *J. Eur. Ceram. Soc.* 45 (3) (2025) 117047.
- [31] D.S.B. Heidary, M. Lanagan, C.A. Randall, Contrasting energy efficiency in various ceramic sintering processes, *J. Eur. Ceram. Soc.* 38 (4) (2018) 1018–1029.
- [32] G. Heiken, D. Vaniman, B.M. French, *Lunar Sourcebook: A User's Guide to the Moon*, no. 1259, Cup Archive, 1991.
- [33] M. Chapman, *The Geology of Mars: Evidence from Earth-Based Analogs*, vol. 5, Cambridge University Press, 2007.
- [34] J.F. Mustard, S.L. Murchie, S. Pelkey, B. Ehlmann, R. Milliken, J.A. Grant, et al., Hydrated silicate minerals on Mars observed by the Mars Reconnaissance Orbiter CRISM instrument, *Nature* 454 (7202) (2008) 305–309.
- [35] M. Elnashar, *Biopolymers, BoD-Books on Demand*, 2010.
- [36] I. Chang, J. Im, A.K. Prasadhi, G.-C. Cho, Effects of Xanthan gum biopolymer on soil strengthening, *Constr. Build. Mater.* 74 (2015) 65–72.
- [37] I. Bozyigit, A. Javadi, S. Altun, Strength properties of xanthan gum and guar gum treated kaolin at different water contents, *J. Rock Mech. Geotech. Eng.* 13 (5) (2021) 1160–1172.
- [38] N. Shiwei, S. Dritsas, J.G. Fernandez, Martian biolith: A bioinspired regolith composite for closed-loop extraterrestrial manufacturing, *PLoS One* 15 (9) (2020) e0238606.
- [39] A.A.B. Moghal, K.V. Vydehi, State-of-the-art review on efficacy of xanthan gum and guar gum inclusion on the engineering behavior of soils, *Innov. Infrastruct. Solutions* 6 (2021) 1–14.
- [40] A. Soldo, M. Miletić, M.L. Auad, Biopolymers as a sustainable solution for the enhancement of soil mechanical properties, *Sci. Rep.* 10 (1) (2020) 267.
- [41] S. Muguda, S.J. Booth, P.N. Hughes, C.E. Augarde, C. Perlot, A.W. Bruno, et al., Mechanical properties of biopolymer-stabilised soil-based construction materials, *Geotechnique Lett.* 7 (4) (2017) 309–314.
- [42] H. Roedel, M. Lepech, D. Loftus, Protein-regolith composites for space construction, in: *Earth and Space 2014*, 2014, pp. 291–300.
- [43] I. Rosa, M. Lepech, D. Loftus, Multiscale modeling and testing of protein-bound regolith and soils, in: *16th Biennial International Conference on Engineering, Science, Construction, and Operations in Challenging Environments*, American Society of Civil Engineers Reston, VA, 2018, pp. 580–590.
- [44] I. Rosa, A. Coto, M. Allende, M. Lepech, D. Loftus, Designing biopolymer-bound regolith composites for maximum compressive strength, in: *Earth and Space 2021*, 2021, pp. 200–214.
- [45] C. Zhou, Y. Gao, Y. Zhou, W. She, Y. Shi, L. Ding, et al., Properties and characteristics of regolith-based materials for extraterrestrial construction, *Engineering* (2024).
- [46] S.J. Dyke, A. Sharma, E.M. Mount, A. Bobet, J.A. Ramirez, Establishing standards for lunar in situ resource utilization structural materials, *AIAA J.* 62 (7) (2024) 2414–2423.
- [47] J. Schmid, V. Sieber, B. Rehm, Bacterial exopolysaccharides: biosynthesis pathways and engineering strategies, *Front. Microbiol.* 6 (2015) 496.
- [48] N. Zhang, X. Li, J. Ye, Y. Yang, Y. Huang, X. Zhang, et al., Effect of gellan gum and xanthan gum synergistic interactions and plasticizers on physical properties of plant-based enteric polymer films, *Polymers* 12 (1) (2020) 121.
- [49] J. Wan, F. Ouyang, H. Xiao, L. Wang, G. Tao, Experimental study on the physical and mechanical properties of modified clay using Xanthan gum and guar gum composite materials, *Sustainability* 16 (13) (2024) 5432.
- [50] G. Hondros, The evaluation of Poisson's ratio and the modulus of materials of a low tensile resistance by the Brazilian (indirect tensile) test with particular reference to concrete, *Aust. J. Appl. Sci.* 10 (1959) 243–264.
- [51] J. Blaber, B. Adair, A. Antoniou, Ncorr: open-source 2D digital image correlation matlab software, *Exp. Mech.* 55 (6) (2015) 1105–1122.
- [52] B. Bresson, F. Meducin, H. Zanni, C. Noik, Hydration of tricalcium silicate (c 3 s) at high temperature and high pressure, *J. Mater. Sci.* 37 (2002) 5355–5365.
- [53] A. Puchalski, A. Hul, J. Nie, T.K. Pietrzak, P. Koblinski, Structure and thermal conductivity of high-pressure-treated silica glass: a molecular dynamics study, *J. Appl. Phys.* 135 (9) (2024).
- [54] J.P. Martin, Decomposition and binding action of polysaccharides in soil, *Soil Biol. Biochem.* 3 (1) (1971) 33–41.
- [55] C. Chenu, Clay—or sand—polysaccharide associations as models for the interface between micro-organisms and soil: water related properties and microstructure, in: *Soil Structure/Soil Biota Interrelationships*, Elsevier, 1993, pp. 143–156.
- [56] E.R. Sujatha, S. Saisree, Geotechnical behaviour of guar gum-treated soil, *Soils Found.* 59 (6) (2019) 2155–2166.
- [57] D. Jussen, S. Sharma, J.K. Carson, K.L. Pickering, Preparation and tensile properties of guar gum hydrogel films, *Polym. Polym. Compos.* 28 (3) (2020) 180–186.
- [58] U.S. Pillai, Discussion of “design of reinforced concrete spandrel beams”, *J. Struct. Div.* 101 (5) (1975) 1151–1153.

- [59] I. Agapkin, E. Slyuta, A review of selective laser sintering/melting techniques for lunar construction, *Acta Astronaut.* (2025).
- [60] M.A. Adajar, J. Tan, A. Adriano, S.B.D. Vera, J.V. Manabat, H. Navarro, Agar biopolymer as a sustainable alternative binder to enhance the strength of low-plasticity soil, *Polymers* 17 (9) (2025) 1253.
- [61] J. Christ, T. Engelsens, E.E. Thybring, L.M. Ottosen, H. Koss, Screening of natural polymers as binder in concrete composites, *Constr. Build. Mater.* 441 (2024) 137405.
- [62] J. Liu, W. Che, X. Lan, M. Hu, M. Qi, Z. Song, et al., Performance and mechanism of a novel biopolymer binder for clayey soil stabilization: Mechanical properties and microstructure characteristics, *Transp. Geotech.* 42 (2023) 101044.
- [63] T. Gualtieri, A. Bandyopadhyay, Compressive deformation of porous lunar regolith, *Mater. Lett.* 143 (2015) 276–278.
- [64] P.J. Collins, J. Edmunson, M. Fiske, A. Radlińska, Materials characterization of various lunar regolith simulants for use in geopolymer lunar concrete, *Adv. Space Res.* 69 (11) (2022) 3941–3951.
- [65] W. Zheng, G. Qiao, Microstructure, thermophysical, and mechanical properties of bulk glass prepared from molten lunar regolith simulant, *Adv. Space Res.* 69 (8) (2022) 3130–3139.
- [66] S.L. Taylor, A.E. Jakus, K.D. Koube, A.J. Ibeh, N.R. Geisendorfer, R.N. Shah, et al., Sintering of micro-trusses created by extrusion-3D-printing of lunar regolith inks, *Acta Astronaut.* 143 (2018) 1–8.
- [67] J. Osio-Norgaard, A.C. Hayes, G.L. Whiting, Sintering of 3D printable simulated lunar regolith magnesium oxychloride cements, *Acta Astronaut.* 183 (2021) 227–232.